20.7.26

EXP NO. 1 - BASIC NETWORKING COMMANDS

AIM:

To study of various network commands used in Linux and Windows.

COMMANDS:

1. arp -a : It will show the IP address of your computer along with the IP address and MAC address of your router.

OUTPUT:

? (172.16.9.212) at 7C:57:58:38:da:a3 [ether] on enp3s0

? (172.16.8.186) at 7C:57:58:38:eb:24 [ether] on enp3s0

? (172.16.8.249) at 00:27:0e:13:f5:77 [ether] on enp3s0

? (172.16.11.207) at 7C:57:58:38:46:87 [ether] on enp3s0

? (172.16.8.140) at 00:27:0e:13:eb:3f [ether] on enp3s0

gateway (172.16.8.1) at 7C:5a:1C:Cf:be:45 [ether] on enp3s0.

2. hostname : It simply displays the name of your computer.

OUTPUT:

localhost.localdomain

3. netstat : It displays a variety of statistics about a computers active TCP/IP connections.

OUTPUT:

Active Internet Connections (w/o servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	localhost.localdoma:ssh	172.16.8.135	Estab
tcp	0	0	localhost.localdom:ssh	172.16.11.207.5919	Estab

4. nslookup : It is used to display DNS details, such as the IP address of a particular computer, the MX records for a dom or the NS servers of a domain.

Eg: nslookup www.instagram.com

OUTPUT:

Server: 172.16.8.1

Address: 172.16.8.1#53

Non-authorative answer:

www.instagram.com canonical name = z-p42-instagram.c10r.instagram.com

Name: z-p42-instagram.c10r.instagram.com

Address: 57.144.56.34

5. ping : best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to other devices.

Eg: ping localhost

OUTPUT:

64 bytes from localhost (::1) : icmp_seq=1 ttl=64 time=0.084ms
64 bytes from localhost (::1): icmp_seq=2 ttl=64 time=0.046ms
64 bytes from localhost (::1): icmp_seq=3 ttl=64 time=0.053ms
64 bytes from localhost (::1): icmp_seq=4 ttl=64 time=0.037ms

6. route : route command is used to show / manipulate the IP routing table. It is primarily used to setup static routes to specific host or networkes.

OUTPUT:

Kernel IP routing table

Destination	Gateway	Genmark	Flags	Metrics	Ref	use
default	gateway	0.0.0.0	UG	100	0	0
172.16.8.0	0.0.0.0	255.255.252.0	U	100	0	0

7 ip : show IP address information assigned to an interface $ ip address show

OUTPUT:

1: lo : <LOOPBACK>, P, LOWER UP > mtu 65536 qd
noqueue state UNKNOWN group default
qlen 1000 link/loopback 00:00:00:00:00 brd
00:00:00:00:00 inet 127.0.0.1/8 scope host lo
valid_lft forever; preferred_lft forever
inet 6 ::1/128 scope host
valid-lft forever preferred-lft forever.

8. ifconfig : Used for configuring and troubleshoo network

OUTPUT: $ ifconfig

enp2s0 : flags = 4163 <UP, BROADCAST, RUNNING MULTICAST>

mtu 1500

inet 172.16.9.200 network 255.255.255.

broadcast 172.16.91.255

inet 6 fe80 ::eb33:8167:72fc prefix 64

0x20 <link>

ether 50:9a:4C:37:fa:C9 txqueuelen(

Rx packets 131162 bytes 19561561 (18.6 MB)

TX packets 2185 bytes 315767 (308.8 KB)

9. mtr : shows you the statistics, including each hop (hostname) with time and loss %.

$ mtr snapchat.com

OUTPUT:

Host	Loss%	Snt	Last	Avg	Best	Wrst	StDe
132.97.163.129	14.3%	46	22.3	12.1	1.9	27.6	6.1
172.31.5.63	3.2%	126	5.2	14.9	3.3	28.3	6.3
209.85.175.48	2.7%	148	21.8	15.4	4.0	30.9	6.4

10. tcpdump : Capturing and displaying packets

$ tcpdump -D

OUTPUT:

1. enp2s0 [UP, Running, Connected]
2. any (Pseudo-device that captures on all interface)
3. lo [UP, Wireless, Not associated]
4. wlp3s0 [UP, Running, Connected]
5. usbmon2 [Raw USB traffic, bus number 2]

RESULT:

Thus, the basic networking commands are executed and verified successfully.